Four years of the Switched Multi-megabit Data Service

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Over the last four years BT's Switched Multi-megabit Data Service has been a major success for the company. This has been due to a unique combination of features that make it genuinely easy for customers to dimension, manage and grow their wide area data networks cost effectively. This paper outlines how the underlying technology and service performance supports a set of features which translate into major business benefits and contribute to competitive advantage. The technology evolution path for SMDS in an ATM-based multi-service environment is described and service interworking scenarios discussed.

1. Introduction

BT launched its Switched Multi-megabit Data Service (SMDS) in the UK in December 1993. The first customer — UKERNA — used the service to provide broadband communications for the academic community's SuperJANET network. Since then, over 180 organisations have chosen SMDS as a key component in their data communications infrastructures.

The service has been taken up by a wide variety of customers, including a third of FTSE-100 companies. All sectors of the UK economy are represented in the customer base — from financial institutions to transport operators, from Internet service providers to police forces, from industrial manufacturers to high street retailers. The sizes of customers' networks range from as few as 3 sites to over 400.

In 1996/97 SMDS volumes rose by 150%. Existing users continued to extend the service to more of their sites, and 70 new customer networks were rolled out. Over 2500 connections are now in service.

2. Service features

Bellcore defined SMDS [1] for the USA in 1989 and this was then adapted to the European Connectionless Broadband Data Service (CBDS) by the European Telecommunications Standards Institute (ETSI). BT's SMDS is based on the Bellcore specifications and those of the European SMDS Interest Group (ESIG) — a particular implementation of the ETSI standard of CBDS.

SMDS has a unique combination of features that make the service a cost-effective way for organisations to satisfy their increasing needs for reliable data communications bandwidth between their sites. With constant change now a fact of business life, SMDS makes it genuinely easy for customers to design, manage and expand their wide area networks.

2.1 Connectionless, switched and high speed

Just like a local area network, SMDS is connectionless — each data packet is examined to see where it should be sent, then routed to that destination as fast as possible (rather than a dedicated call for each transfer as in a connection-oriented service).

Unlike most LANs, SMDS is also switched — equipment in the network determines the transmission path for each packet.

With BT's SMDS, access classes are currently available at 0.5 Mbit/s, 2 Mbit/s, 4 Mbit/s, 10 Mbit/s, 16 Mbit/s and 25 Mbit/s and the chosen bandwidth is always available. On the higher speeds (4 Mbit/s and above) a 'credit manager' allows a burst capability up to 25 Mbit/s. A 0.2 Mbit/s access class is being launched in January 1998.

This feature combination means that SMDS customers benefit from the economies of sharing a national broadband transmission infrastructure, and get real-time 'any-to-any' connectivity between sites of widely varying bandwidth needs, without having to dimension all the permutations of their traffic routes.

2.2 Secure and flexible addressing

SMDS uses the global E.164 addressing standard. Each access is assigned a unique 15-digit address (held by the network equipment), which identifies that line.

Each SMDS access also has customer-defined address 'screens' (again, held within the network), which determine the SMDS addresses from which that customer is allowed to receive data and to which they can send data. The network checks each packet against these 'source' and 'destination' screens, and discards packets if they do not meet the criteria.

Customers can therefore maintain secure closed user groups (e.g. for intranets), which can be readily extended to their business partners such as suppliers and customers.

2.3 Managed

The SMDS service is delivered over a network that is inherently resilient, using duplicated transmission equipment and a highly meshed network design. Round-the-clock monitoring of the network ensures that any faults are isolated, and repaired rapidly, with minimal disruption to customer traffic. Furthermore, each access is proactively monitored by BT, to check for any faults on that access line.

Thus, SMDS customers get the in-built resilience of a public network, and fast repair of access faults.

2.4 Flat-rate prices

BT's SMDS now has 36 service points throughout the UK. Each has a 25 km radius, within which the service is charged at rates which are completely independent of distance.

With most access classes customers pay a connection charge and an access rental, with no additional usage-related charges, and no route-related charges. Usage-based charging has recently been introduced for the 10 Mbit/s access class.

This means that SMDS customers can budget for predictable service charges, which become increasingly attractive as the number of sites increases, and as the distance between sites increases.

3. Service infrastructure

The service features described in section 2 are provided over a platform which operates the SMDS protocol on a meshed network.

3.1 The SMDS protocol

SMDS exchanges variable-length packets which can be up to 9188 octets of information. The protocol operating over the SMDS network interface (SNI) is the SMDS interface protocol (SIP) and has three levels (Fig 1).

Firstly, the user information has a header and trailer attached to form the level 3 protocol data unit (PDU). The header contains the ITU-T E.164 source and destination addresses, as well as various tags and identifier fields. The trailer contains similar identifiers and a checksum. These level 3 PDU addresses establish the connectionless packet service and enable the level 3 PDU to be routed through the network without requiring call set-up or special provisioning.

The level 3 PDU is then divided into the level 2 PDUs (or cells), each 53 bytes in length, which are transmitted across the SNI using the SIP. The SNI can be assigned up to 16 individual E.164 addresses. Level 2 corresponds to the ATM layer in the B-ISDN recommendations.

Level 1 is the G.703 digital physical interface. The 0.2 Mbit/s, 0.5 Mbit/s and 2 Mbit/s access classes are delivered on an E1 (2 Mbit/s) line to the switch and the 4, 10, 16 and 25 Mbit/s on an E3 (34 Mbit/s) line.

3.2 Layer 3 protocol support

SMDS supports a wide range of layer 3 protocols, e.g. TCP/IP, Novell, IPX, SNA, DECnet, X.25, VINES and

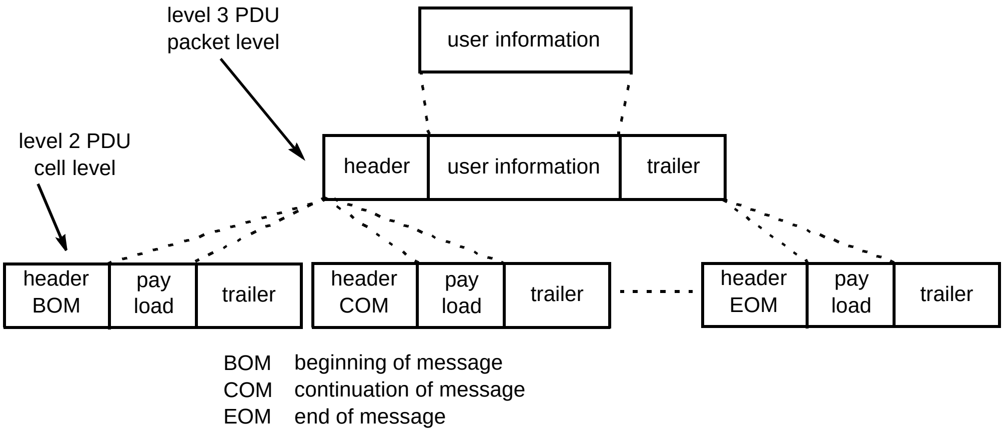

Fig 1 The SMDS protocol.

Appletalk. Many of these are used by the current customer base with TCP/IP being predominant.

3.3 Core technology

SMDS is currently delivered on a network of GPT/ Siemens EWSM cell switches which use the distributed queue dual bus (DQDB) protocol (IEE802.6). A central mesh of the switches forms the network core and an outer layer provisions customer accesses. Alcatel-supplied Cascade Communications concentrators (CBDX-9000) provide the access mechanism for low-speed SMDS (0.5 and 2 Mbit/s).

3.4 Access technology

High-speed SMDS access

Connection to the SMDS service for access classes of 4 Mbit/s and above, is provisioned on an optical fibre line which connects into the SMDS customer premises equipment (CPE) (Fig 2). This SMDS CPE is a line terminating card and a data service unit or DSU.

This then connects into the customer router via a standardised interface — the data exchange interface (DXI version 3.2), which can be either a V.35 (for the 4 Mbit/s access class) or high-speed serial interface (HSSI) (for access classes 4 Mbit/s and above). The service can be managed inclusive or exclusive of the customer router — the router itself provisioned either by the customer or from BT's portfolio. The managed router service option is offered through BT Sycordia Solutions.

Low-speed SMDS access

Low-speed SMDS provides most of the standard features of SMDS, such as any-to-any connectivity, address screening, and multicasting; however, the ability to burst above the subscribed access class is not provided. Access rates for the low-speed SMDS service are 0.5 Mbit/s (448 kbit/s), 2 Mbit/s (1.92 Mbit/s) and 0.2 Mbit/s (available from January 1998). A low-speed switch is necessary to concentrate all the accesses to a single, large pipe into the SMDS network (Fig 3).

A typical environment where low-speed SMDS is the ideal internetworking technology is where many branch offices have a need to communicate with each other as well as a small number of main sites (regional offices or data centres). For example, in the banking industry the branch sites would typically be connected via low-speed SMDS into a core of the main sites which have high-speed SMDS accesses.

3.5 Network and service management

SMDS is supported by four BT functional areas, the Customer Service Centre (CSC), Advanced Services Technical Centre (ASTC), Network Management Centre (NMC, also known as the Network Operations Unit (NOU)), and BT Laboratories (BTL):

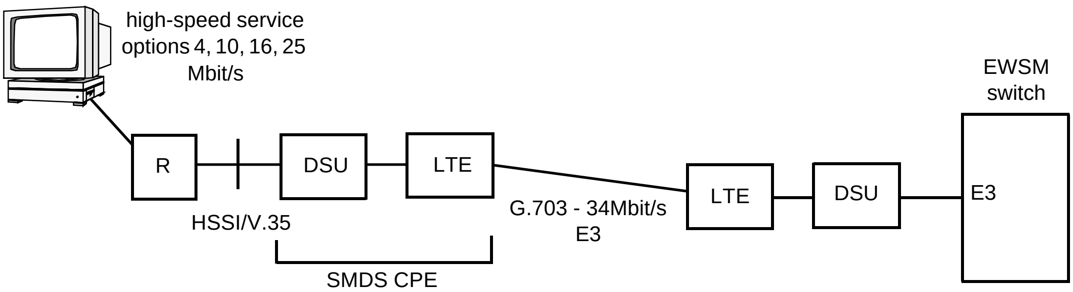

Fig 2 High-speed SMDS implementation.

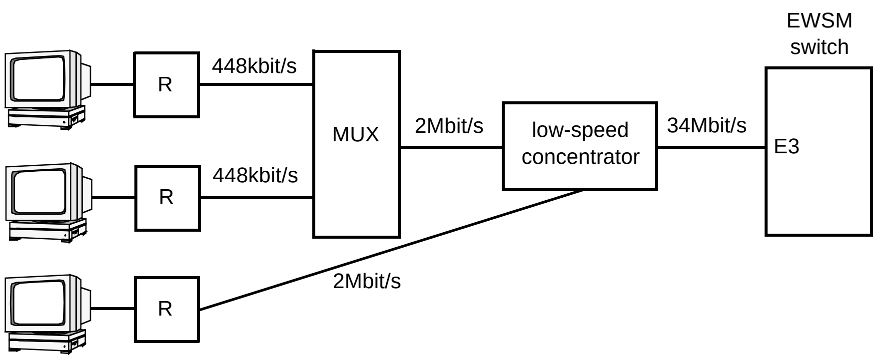

Fig 3 Low-speed SMDS implementation.

- Customer Service Centre — the front office for the provision and maintenance of major customer BT services,
- Advanced Services Technical Centres — ASTCs act as technical support and helpdesk for all the 'advanced services', such as FBS, SMDS, ATM, and short-haul data services, and operate in the service management domain logging faults and undertaking primary diagnostics,
- Network Management Centre/Network Operations Unit — the NMC for SMDS, which is situated at Walsall, undertakes detailed diagnostics and despatches field engineers,
- BT Laboratories undertakes detailed technical consultancy and customer network simulations.

4. SMDS customers, business applications and benefits

To understand the benefits of using SMDS, it is important to put the service into the context of the customer's business. Any organisation performs a wide variety of activities that, taken together, deliver 'value' to its own customers. These activities can be represented as a 'value chain' of tasks required to deliver and service the organisation's products, and of support activities [2] (see Fig 4).

Each type of activity in the chain is a potential contributor to an organisation's competitive advantage in an industry, either by creating a cost advantage relative to its competitors or by differentiating the organisation's product in the eyes of its customers.

All parts of the value chain are present to some extent in all organisations; however, depending on the industry, certain activities may be particularly vital to competitive advantage. For instance, research and development activities are especially important in the pharmaceuticals industry, while branding and promotion may be the key differentiator for companies selling soap powder.

As far as this discussion is concerned, the key point is that each and every value-creating activity uses and generates **data** in various forms. With information technology revolutionising the role of data in many business activities, the ability to transfer information easily into, within and out from the organisation takes on strategic importance. In particular, the scope and capabilities of a wide area network (WAN) can make a profound impact throughout an organisation's value chain and, as a consequence, on its competitive advantage.

BT's SMDS is currently the dominant broadband data WAN solution in the UK. The service has been used by many customers to 'change the way they work'.

4.1 Inbound logistics

Inbound logistics covers activities such as scheduling, receiving, checking and storing new supplies of material used in the production of the organisation's product.

In a manufacturing business the importance of such activities is fairly clear; 'just-in-time' supply processes used in the car industry, for example, have made a significant impact on cost structures. In a service business, however, the role of inbound logistics is more subtle, but can be just as important.

For instance, the National Blood Service's (NBS's) core operation is the collection and distribution of blood and blood products. Each year the NBS collects around 2.5 million units of blood, from nearly 2 million donors. The blood is transported to one of 14 regional blood centres for

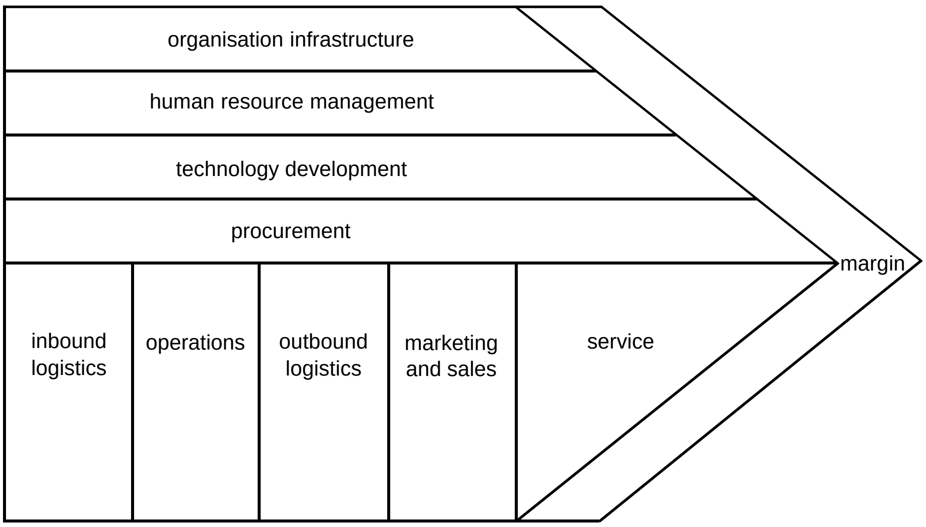

Fig 4 The generic value chain [2]. [Diagram courtesy of The Free Press]

processing into three main components — red cells, plasma and platelets — which are then distributed to hospitals and laboratories. In such a business the recruitment of donors and management of information about donors is clearly extremely important.

NBS chose BT's SMDS to link the blood centres with three data centres. This enabled the national roll-out of NBS's 'PULSE' application, which has modules for donor management, production and a donor marketing database of 5-6 million people. The any-to-any nature of SMDS means that, while for 95% of the time a blood centre will access its nearest data centre, it can access **any** of the three data centres at any time to get the information it needs. As a result of improved access to data, the NBS has seen huge benefits — increased ability to supply blood products, reduced costs of donor testing, no loss of donor history, and the ability to monitor peaks and troughs in the national blood stock.

4.2 Operations

Operations includes the activities associated with actually creating the organisation's final product. For a manufacturer, this may typically include machining, assembly, testing, and packaging.

An example of a service organisation using BT's SMDS at the core of its operations is BACS Ltd, which operates inter-bank networking services in the UK on behalf of the Cheque and Credit Clearance Company (C&CCC). Under the previous system for cheque clearance, a single cheque was read by the payer's bank **and** by the bank of the payee — effectively a double procedure. The relevant information would be read by machines on to magnetic tape, which was then taken by courier from one bank's computer centre to the other. The process was error-prone, laborious and expensive. BACS automated the process by implementing an SMDS network, which was connected to each of the twelve major clearing banks. The benefits have been a new line of business for BACS, improved reliability, and lower cost of cheque clearance for C&CCC.

4.3 Outbound logistics

Outbound logistics covers activities associated with distributing the organisation's product to its customers.

Intelligent use of WANs in the distribution channel has been shown to deliver superior customer service. Nissan UK, for example, runs a data network which gives dealers on-line access to important information such as vehicle availability, warranties, paint charts, etc. The core of the network is based on BT's SMDS, linking sites such as Nissan's UK headquarters, European data centre, training centre, national distribution centre and certain regional offices. The links out to the dealers themselves are currently ISDN. SMDS provided a cost-effective and scalable way of increasing core network capacity. The dealers are noticing increased reliability, applications are running faster and more comprehensive product information can be provided.

4.4 Sales and marketing

Sales and marketing includes customer account management, requirements capture, pricing, promotion, and channel management.

Many promotional marketing activities, for example, involve the sharing of large amounts of data between the organisation and their external agencies. BT has itself recently revolutionised the way it runs its own programme for promoting products and services to residential customers. By using SMDS to connect key BT sites with those of external consultants and agencies the lead times to plan and implement campaigns are being drastically reduced, and the quality of the process significantly improved. SMDS's addressing features were ideal for setting up the secure yet flexible closed user groups that such an application required, and the ability to interconnect with MCI's SMDS service allowed sites in the US to be seamlessly incorporated into the network.

4.5 Service

Service covers those post-sales activities which are required to maintain the value of the product or service that has been (or, in the case of a service, is being) delivered to customers. This would typically be dominated by problem-handling processes and by mechanisms for supporting customers with information about the product they have purchased.

In one major UK housing charity, for instance, SMDS provides instant access for the customer service representatives to the data they need to answer customer queries.

4.6 Procurement

Procurement covers the activities associated with purchasing any externally provided products and services used by the organisation. This is a function (not necessarily contained within a single organisational unit) that spans all areas of an organisation's activities. The purchases may range from the raw materials used in a manufacturing process, to the PCs used in the organisation's offices, to the market research services used in the pricing of the organisation's product. In some industries — clothing retail, for example — an organisation's competitive advantage could hinge on how well it handles certain aspects of its procurement activities.

One of the world's leading car manufacturers recently recognised the benefits of broadband communications with its many suppliers. Some of these were already using SMDS for their own purposes. A simple re-configuration of address screens allowed these companies to communicate securely within their own 'community of interest'.

4.7 Technology development

Technology development covers those activities concerned with improving the organisation's product set, and improving the processes used in creating and delivering the product. In this context, 'technology' has a broad scope — it includes operational procedures as well as product design and manufacturing techniques. Like procurement, technology development spans all the organisation's activities in some form or another.

It has already been mentioned that technology development is particularly important to competitive advantage in the pharmaceuticals industry. To assist in achieving economies of scale following a merger, a leading pharmaceuticals group used BT's SMDS to facilitate close working between its two, previously separate, product development organisations. The relative ease of dimensioning a network based on a connectionless service helped to ensure a rapid roll-out once the merger had gone through, while the bursty bandwidth that SMDS offers was particularly suited to the fast transfer of large files of research data and drug designs.

4.8 Human resource management

Human resource management covers activities such as recruitment, training and pay of the organisation's personnel.

Knowledge industry professions such as management consultancy are critically dependent on the skills of their people, and on their continual development. Several major consultancy firms use BT's SMDS for their own intranets, which, among other things, support personnel systems, and provide access to training material.

4.9 Organisation infrastructure

Finally, organisation infrastructure covers activities that support the entire value chain, such as financing, accounting, general management, planning, project management, and relations with investors and regulators. As with all other parts of the value chain, this area is a potential source of competitive advantage in certain situations, for instance, rigorous financial control could be an important factor in an industrial conglomerate's success.

The Civil Aviation Authority (CAA) reviewed its business data network when a study in 1994 revealed that the existing WAN lacked the flexibility and scalability the CAA would need to cope with the increasing take-up of IT. For instance, new client/server applications for finance and personnel were about to go live, and the existing network could not cope. The CAA decided to implement an SMDS network to link its six major sites throughout the UK, and soon found that the service significantly enhanced the speed and flexibility of the organisation's business communications. A key benefit was the ease of adding further access to their SMDS virtual private network, as the bandwidth requirements at other sites grew.

5. Service performance

5.1 Latency

All data packets traversing a communications network experience a distance-dependent transmission delay which is a function of packet size. It is important that this latency is minimised, from both an application and protocol perspective. Certain application types are particularly sensitive to delay, for example, in the client/server environment and with Novell IPX applications. Layer 3 protocol performance can also be a function of wide area latency, but can be optimised, e.g. using window sizing.

The latency of SMDS is inherently low and service level objectives are provided to customers.

Figure 5 shows a plot of SMDS latency for both a high-speed to high-speed SMDS (10 Mbit/s) connection and a low-speed to low-speed SMDS (2 Mbit/s) connection. These are actual measured times across a live SMDS network. It can be seen that the round-trip time is substantially less on the high-speed connections, with little variation across the whole range of packet sizes, whereas the low-speed connection latency is more a function of the packet size.

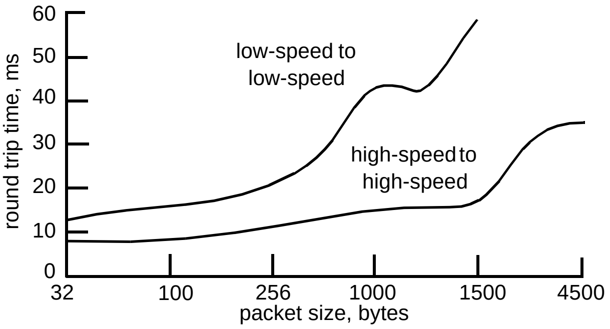

Fig 5 Measured round-trip latencies for SMDS.

The round-trip time (or latency) across the network has a direct relationship to the throughput of data between sites.

5.2 Throughput

Throughput is a measure of the successful transport of packets across the network as a function of time. The upper throughput limitation is the bandwidth available at the lowest access class point in the virtual private network. Therefore, a 1.92 Mbit/s SMDS site sending data to a 448 Kbit/s SMDS site could achieve a maximum throughput of 448 Kbit/s.

Throughput can also, however, be affected by CPE (i.e. router) performance, e.g. buffer sizing and non-optimal protocol performance combined with latency.

For example, a single TCP/IP session across an SMDS network with a round trip delay of 40 ms and a TCP window size of 4000 bytes would have a maximum throughput of:

$$\frac{\text{window size, bits}}{\text{round-trip delay, secs}} = \text{throughput, bit/s}$$

$$\frac{(4000 \text{ bytes} \times 8 \text{ bits/byte})}{0.04 \text{ secs}} = 800 \text{ kbit/s}$$

Obviously, this does not necessarily make good use of the available SMDS bandwidth. In this case improved throughput may be achieved by increasing the window size of the TCP protocol; also, the example is for a single session. A much better bandwidth utilisation can be achieved with multiple parallel sessions.

5.3 Packet loss

Packet loss can occur as a result of congestion in the network or an equipment fault. The SMDS network is engineered and monitored to ensure that the service objectives are met on an ongoing basis. The objective is to have a packet loss of better than 1 in every 10 000 data packets transmitted. Full details are available in the SMDS service description [3].

5.4 Service availability

The SMDS network core is extremely resilient as it operates with dual trunk connections between switch sites and in the event of a failure on one, the traffic can be switched to the other in seconds. The switches in the network have built in redundancy with dual cards and power supplies. Faults are addressed quickly, providing SMDS with a high level of availability specified at 99.9% per access.

When a fault occurs, it is important to the customer that it is fixed rapidly. SMDS incorporates a level of service, aiming to repair faults within 5 hours (24 hours a day, 365 days of the year). In 1996, an average of 98.3% faults were cleared within 5 hours — well in excess of the target of 92%.

BT also operates the SMDS service level scheme in which BT will reduce the annual rental on an access line if its availability measured over 12 months falls below the 99.9% target. The reduction is a percentage of the annual rental price, and ranges from customers getting 5% discount with a service availability of less than 99.9% but more than 95%, to a full 100% discount if the service availability drops to 75% average over the year.

6. Service evolution

To take advantage of the economies of scale of a reusable network platform, BT plans to migrate several existing services on to an ATM-based multiservice platform over the next few years [4]. One of the first such services in the UK will be SMDS. This provides several challenges for the network designer, including how to provision:

- SMDS supported on a core ATM network,
- SMDS delivered on an integrated ATM access to the customer,
- interworking of existing SMDS and future integrated access SMDS sites.

6.1 SMDS supported on a core ATM network

Support of an SMDS service over a core ATM network is transparent to the SMDS customer, but requires transmission equipment to interface the existing DQDB-based network [5] to the new infrastructure.

Trials of the core ATM transport of SMDS were successfully completed during the summer of 1997 using the same platform as the BT CellStream service. Gateway switches were interfaced to CellStream via E3 ATM UNIs, and the trial was used to investigate issues of ATM core trunk configuration, traffic management and resilience.

6.2 SMDS on an integrated ATM access

BT is studying the provision of SMDS services via an integrated access mechanism and a main part of any such solution is an ATM multiplexer (mux) at the customer premises. The purpose of the ATM mux is to multiplex (mux) all customer traffic service streams together over an ATM UNI/NNI and to provide support for existing service interfaces. This ATM link would then provide the boundary between the customer premises and the public ATM network.

There are two different types of service mux which could be used — a multiservice mux or a pure ATM-based mux.

The multiservice mux (see Fig 6) would provide interfaces which can interconnect with all existing customer services (SMDS DXI, Frame Relay, etc). It would provide adaptation into ATM for each service stream using the appropriate AAL, and would multiplex the different service streams together over an ATM UNI/NNI.

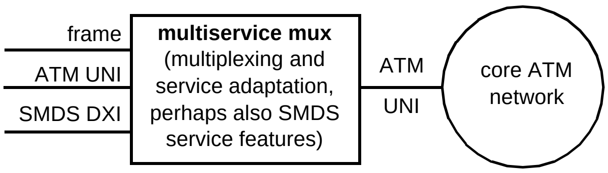

Fig 6 The multiservice multiplexer.

This approach has the advantage that the customer does not need to modify existing CPE to provide transport over an ATM access and the service mux may also be able to support some of the SMDS service features such as source address validation and destination E.164 address to ATM VP/VC mapping. However, the disadvantages are the current lack of service muxes which provide support for SMDS and the fact that the CPE which supports SMDS is likely to be high functionality which may equate to high cost.

The ATM-based mux (see Fig 7) would be a lower functionality piece of equipment which provides no service support and simply enables multiplexing of a number of ATM traffic streams on to an ATM UNI. Each ATM traffic stream would carry a different service.

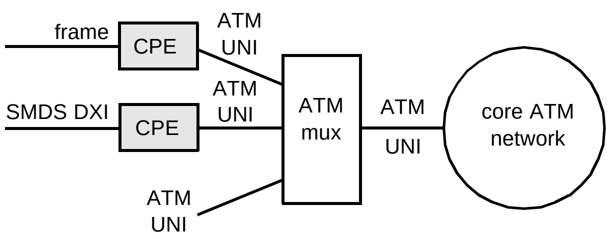

Fig 7 The ATM multiplexer.

The service interface to ATM for legacy services would need to be carried out by the customer CPE, e.g. by an ATM card in an SMDS router. Native ATM services could be interfaced directly to the ATM mux via an ATM UNI. The advantages of this approach are the widespread availability of kit which provides this functionality (since it is basically a small ATM switch). This could make it easier to choose a product to meet the requirements at the right cost. The disadvantages are that it does not offer the easiest upgrade option to customers with a number of legacy service interfaces. A customer site migrating to ATM would not just need an ATM mux but would also be required to upgrade CPE to support ATM functionality.

6.3 *Interworking of legacy and integrated access SMDS*

In 1994 the ATM Forum and SMDS Interest Group co-operated to define a specification for the interworking of SMDS and ATM at layer 3 [6]. Site interworking can currently be supported easily using this specification via routers, e.g. at the IP (Internet protocol) level. However, in this scenario SMDS service features are not available end to end. At the source site, IP is encapsulated in SMDS for wide-area transport to the interworking router. Here the IP packets are retrieved and encapsulated in ATM for wide-area transmission to the destination site. The non-SMDS site does not have access to the full SMDS service feature set, e.g. it cannot set up a multicast from itself to several destination SMDS sites.

There are two ways forward for realising interworking between legacy SMDS sites and future integrated access SMDS sites.

Firstly, a gateway device could be used (see Fig 8). Each integrated SMDS access would have a PVC between it and the gateway device across the ATM core. The gateway would then map these VCs to a DXI interface on the legacy SMDS network, one DXI per VC. The gateway device would need to be able to implement SMDS service features in order to provide true interworking. This approach would most probably be used to satisfy specific customer requirements in the shorter term and where a small number of ATM sites are involved.

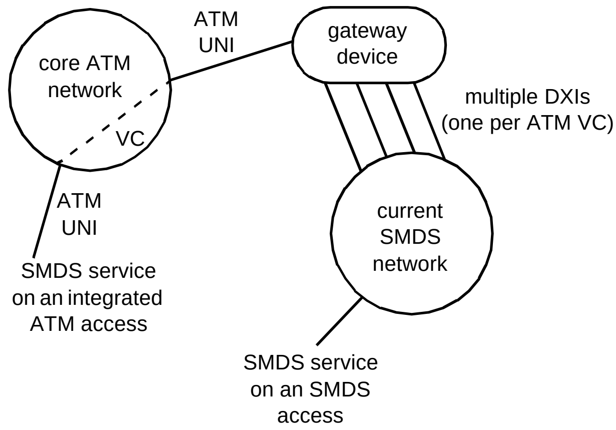

Fig 8 Use of a gateway device for interworking.

The second approach would be to implement a connectionless server (CLS) (see Fig 9). This provides SMDS service features such as address screening, routeing and multicasting. This server could be distributed or in a single location within the core or access network. It may also be, in part, within the CPE. The main advantage of this approach is that SMDS traffic between ATM sites would not go via the legacy SMDS network but via the connectionless servers. The CLS would need to be able to accommodate a large number of ATM interfaces, i.e. one per SMDS DXI mapped to that particular CLS. Between CLSs, a smaller number of VPs could be used. The CLS approach to interworking is more scalable than the gateway device but no fully functional equipment yet exists in the market.

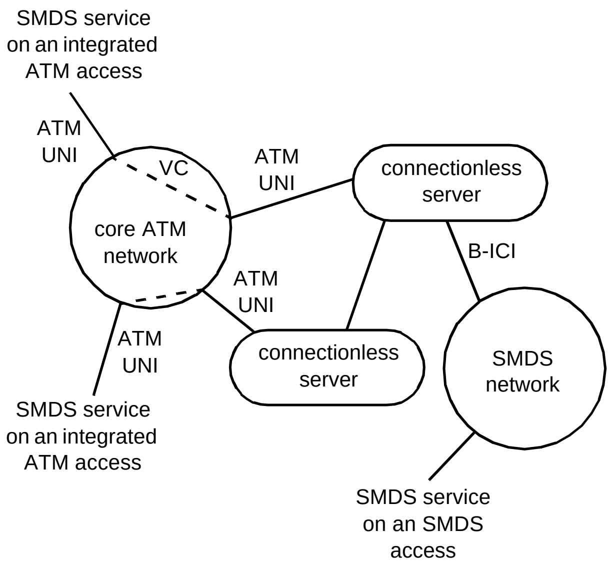

Fig 9 Use of a connectionless server for interworking.

7. Conclusions

Over the last four years BT's switched multi-megabit data service has put the UK at the forefront of broadband data communications. Its success has been due to a unique combination of features that make it genuinely easy for customers to dimension, manage and expand their wide area data networks cost effectively.

This paper has outlined how the underlying technology supports these features which translate into major business benefits for customers. Examples have shown how SMDS can make a positive impact throughout an organisation's value chain, thereby contributing to competitive advantage.

The emergence of ATM technology brings the opportunity for SMDS to be supported on an infrastructure that is shared with other broadband services, thus realising economies of scale. The technology evolution path for SMDS in this multiservice environment is clear and interworking studies are well advanced.

Building on the success of the service over the last four years and as ATM services begin to appear, SMDS will continue into the future to support a large number of corporate customers who require the bandwidth, high levels of performance and scalability that it offers.

Acknowledgments

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Debbie Lewis gained a BSc in Physics from Durham University in 1987 and joined the Submarine Systems section of BT Laboratories in the same year. Here she was in a team providing technical support for the procurement of long-haul transatlantic fibre systems — TAT8, TAT9 and TAT12/13. She gained an MSc in Telecommunications and Information Systems from Essex University in 1989. In 1994 she joined the Broadband and Data Networks unit to work on the SMDS service. She now leads a team providing technical support for the service and network transport testing expertise. She is an Associate Member of the IEE.

David Mack-Smith joined BT in 1990, having graduated with a First in Electrical and Information Sciences at Cambridge University. He spent two years at BT Laboratories researching and developing optical fibre systems for BT's access network, before moving to London to join the Products and Services Management division. Here, he worked on the team which developed and launched the Caller Display and Call Return (1471) services. He then joined the team which developed, launched and managed a remote meter reading service using 'no-ring call' technology. In January 1997, he became product manager for BT's Switched Multi-megabit Data Service.

Brian Poole joined BT Laboratories in 1985 as an apprentice, through which he attained the HNC Electronic Engineering in 1989. Upon finishing the apprenticeship he joined the Advanced Subjective Testing unit where he worked on subjective testing of new products and services until March 1995. During his time there he specialised in acoustic measurement and reproduced sound, and attained the Diploma in Acoustics in 1992. He then joined the Information Visualisation group for a short spell looking at evaluating ATM from an application perspective before moving to the Broadband and Data Networks unit in October 1995. In his role within the unit he gives technical customer support on all aspects of BT's broadband services, and is the main source of customer technical enquiries for the SMDS service. He is currently studying for a BSc (Hons) in Information Technology with the Open University.